

Lecture Notes IV

Course: Functional Analysis

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Topic: The Lebesgue Measure

Proof 1.27 (Additivity if one of the sets is open). If $m^*(G) = \infty$, then the equality is immediate since $m^*(A \cup G) \geq m^*(G)$ implies $m^*(A \cup G) = \infty$. If $m^*(G) < \infty$, then we only need to show that $m^*(A \cup G) \geq m^*(A) + m^*(G)$ because the reverse inequality follows from σ -subadditivity. First we consider $G = (a, b)$ for some $a < b$ in $\mathbb{R}$. Define $\tilde{A} = A \setminus (\{a\} \cup \{b\})$, then $m^*(A) = m^*(\tilde{A})$. Let (I_n) be a countable open cover of $A \cup G$ and for each n define

$$J_n = I_n \cap (-\infty, a), \quad K_n = I_n \cap (a, b), \quad L_n = I_n \cap (b, \infty)$$

so that defining

$$D_n = J_k \quad n = 2k - 1, \quad D_n = L_k \quad n = 2k,$$

we get that (D_n) is a countable open cover of A (recall that $A \cap G = \emptyset$) and (K_n) is an open countable cover of G . Thus

$$\ell(I_n) = \ell(J_n) + \ell(K_n) + \ell(L_n)$$

which in the limit yields

$$\sum_{n \in \mathbb{N}} \ell(I_n) = \sum_{n \in \mathbb{N}} (\ell(J_n) + \ell(L_n)) + \sum_{n \in \mathbb{N}} \ell(K_n) \geq m^*(A) + m^*(G),$$

which in turn implies that $m^*(A \cup G) \geq m^*(A) + m^*(G)$ (recall that (I_n) is a countable open cover of $A \cup G$). By induction, we can conclude that if G is a union of m disjoint opens which themselves are disjoint from A , then $m^*(A \cup G) = m^*(A) + m^*(G)$. If G is any arbitrary open subset with $A \cap G = \emptyset$ (the case $G = \emptyset$ is trivial and should be treated separately, so assume $G \neq \emptyset$), then G is the union of countably many disjoint open intervals I_n in $\mathbb{R}$ (see Sherbert 11.1.9 for a proof), clearly, the intervals are also disjoint from A so that for any given $m \in \mathbb{N}$ we have

$$\bigcup_{n=1}^m I_n \subseteq G \implies A \cup \bigcup_{n=1}^m I_n \subseteq A \cup G \implies m^* \left(A \cup \bigcup_{n=1}^m I_n \right) \leq m^*(A \cup G)$$

but since m^* is additive, we have

$$m^*(A \cup G) \geq m^*(A) + \sum_{n=1}^m m^*(I_n) = m^*(A) + \sum_{n=1}^m \ell(I_n)$$

for all $m \in \mathbb{N}$, and thus taking the limit on m we get

$$m^*(A \cup G) \geq m^*(A) + \sum_{n \in \mathbb{N}} \ell(I_n)$$

so that $m^*(A \cup G) \geq m^*(A) + m^*(G)$ and the proof is done.

Proof 1.30 (Additivity if one of the sets is Borel). Let $\varepsilon > 0$ and let F be closed and such that $F \subseteq B$ and $m^*(B \setminus F) < \varepsilon$, (the existence of such F is justified by Theorem 1.29) so

$$m^*(A \cup B) \geq m^*(A \cup F) = m^*(A) + m^*(F) = m^*(A) + m^*(B) - m^*(B \setminus F) > m^*(A) + m^*(B) - \varepsilon$$

since $\varepsilon > 0$ was arbitrary, the conclusion now readily follows.

Proof 1.32 (Lebesgue outer measure is a measure on $\mathcal{B}$). $m^*(\emptyset) = 0$ is obvious. Suppose (B_n) is a sequence of disjoint measurable sets. Then for any $n \in \mathbb{N}$ we have

$$m^* \left(\bigcup_{k \in \mathbb{N}} B_k \right) \geq m^* \left(\bigcup_{k=1}^n B_k \right) = \sum_{k=1}^n m^*(B_k) \implies m^* \left(\bigcup_{k \in \mathbb{N}} B_k \right) \geq \sum_{k \in \mathbb{N}} m^*(B_k),$$

and the reverse inequality is just σ -subadditivity, i.e., m^* is σ -additive on $\mathcal{B}$.